AnyRecon: Arbitrary-View 3D Reconstruction with Video Diffusion Model

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Research Questions

- How can video diffusion models be adapted to perform 3D reconstruction from arbitrary numbers of unordered sparse input views?
- How can generation and reconstruction be coupled through an explicit 3D geometric memory to maintain consistency across long trajectories?
- How can computational efficiency be maintained while supporting flexible conditioning cardinality for large-scale scenes?

Core Idea

Existing diffusion-based 3D reconstruction methods are limited to conditioning on only 1-2 views, which restricts geometric consistency for large scenes. AnyRecon addresses this by creating a persistent global scene memory that can condition on arbitrary numbers of unordered input views, while maintaining an explicit 3D point cloud that gets iteratively updated as new views are generated. The key insight is coupling generation and reconstruction in a closed loop: newly generated views update the 3D geometry memory, which then guides subsequent view generation through geometry-driven view selection and spatial conditioning.

Key Formula

4-step distillation loss

$$\mathcal{L}_{gen} = \mathbb{E}_{z_t, t}\left[\frac{1}{2}\left|\left|\hat{x}_\theta(z_t) - \mathrm{sg}\left(\hat{x}_\theta(z_t) + \eta\frac{\hat{x}_\psi(z_t) - \hat{x}_\phi(z_t)}{\sigma_{\mathrm{norm}}}\right)\right|\right|_2^2\right]$$

Generator loss for Distribution Matching Distillation where x̂_θ is student prediction, x̂_ψ is teacher prediction, x̂_φ is critic prediction, η is step size, σ_norm is normalization factor, and sg() is stop-gradient operator. This enables 4-step sampling instead of 50 steps for efficiency.

Geometry-driven view selection score

$$s_i = \frac{|\mathcal{V}_{novel} \cap \mathcal{S}_i|}{|\mathcal{V}_{novel}|}$$

Score for selecting conditioning views where V_novel is the set of geometry points visible from target viewpoint and S_i is the subset of points reconstructed from capture view i. This quantifies geometric contribution of each view to current target perspective for retrieval.

Key Figure

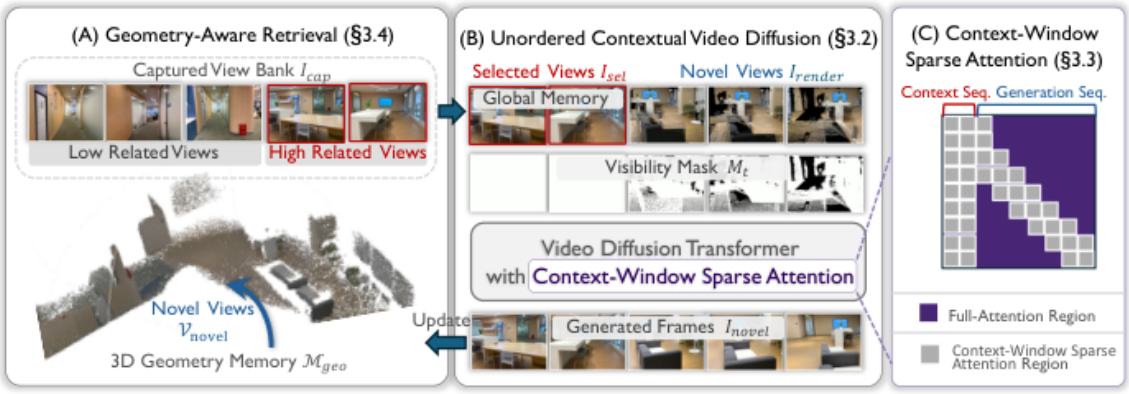

Fig. 2: Pipeline of AnyRecon. Given arbitrary sparse input views organized in a capture view bank $\mathcal{I}_{cap}$, we perform geometry-aware retrieval to select spatially informative views for each novel trajectory segment. The selected views $\mathcal{I}_{sel}$, together with geometry renderings under target viewpoints $\mathcal{I}_{render}$, are fed into a video diffusion transformer equipped with context-window sparse attention for scalable long-range conditioning. The generated novel views are then used to update the 3D geometry memory $\mathcal{M}_{geo}$, forming a closed loop between generation and reconstruction.

3D structures from input images before projection into the diffusion pipeline. Nevertheless, these diffusion model typically condition on a very limited number of captured frames (e.g., only the first and last), which limits their ability to capture global scene context and fine-grained appearance fidelity. In contrast, our AnyRecon utilizes a global scene memory to incorporate an arbitrary number of reference views $\mathcal{I}_{sel}$ and enforces strict spatial alignment through channel-wise concatenation of visibility masks M_t and rendered observations I_{render}, closing the gap between generative priors and explicit 3D geometry.

3 Method

3.1 Overview

We present **AnyRecon**, a framework for sparse-view 3D reconstruction that supports arbitrary and unordered inputs while preserving geometric consistency across long viewpoint changes. Our method alternates between (1) diffusion-based trajectory generation to create more views and (2) geometry refinement using the generated views. This forms a closed loop that progressively reconstructs the scene, enabling scalable processing of long trajectories and large-scale inputs.

As illustrated in Fig. 2, our framework operates in three key stages to form the generation-reconstruction loop. First, *Initial Geometry Construction:* All input views are organized into a captured view bank $\mathcal{I}_{cap}$, from which an initial 3D geometry memory $\mathcal{M}_{geo}$ is established via a feed-forward point map estimation method (e.g., VGGT [19] or π^3 [22]). Second, *Novel View Generation:* To synthesize novel views on user-specified trajectory V_{novel}, we chop the entire trajectory into small for efficiency. For each segment, we perform geometry-aware

Figure 2: Shows the complete AnyRecon pipeline including geometry-aware retrieval, unordered contextual video diffusion, and

context-window sparse attention that enables scalable conditioning on arbitrary numbers of views.

Limitations

- Performance depends on quality of initial 3D geometric memory reconstruction
- Requires basic structural coherence in initial point cloud for effective guidance
- May fail with minimal view overlap leading to insufficient geometric guidance
- Relies on feed-forward geometry estimation models which may introduce artifacts